TIC 251077178

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        0.684402 d
  Epoch (t0)    1411.3340 BTJD
  Duration      0.0840 d
  Depth         2.742 %  (27,417 ppm)
  SNR           319.2

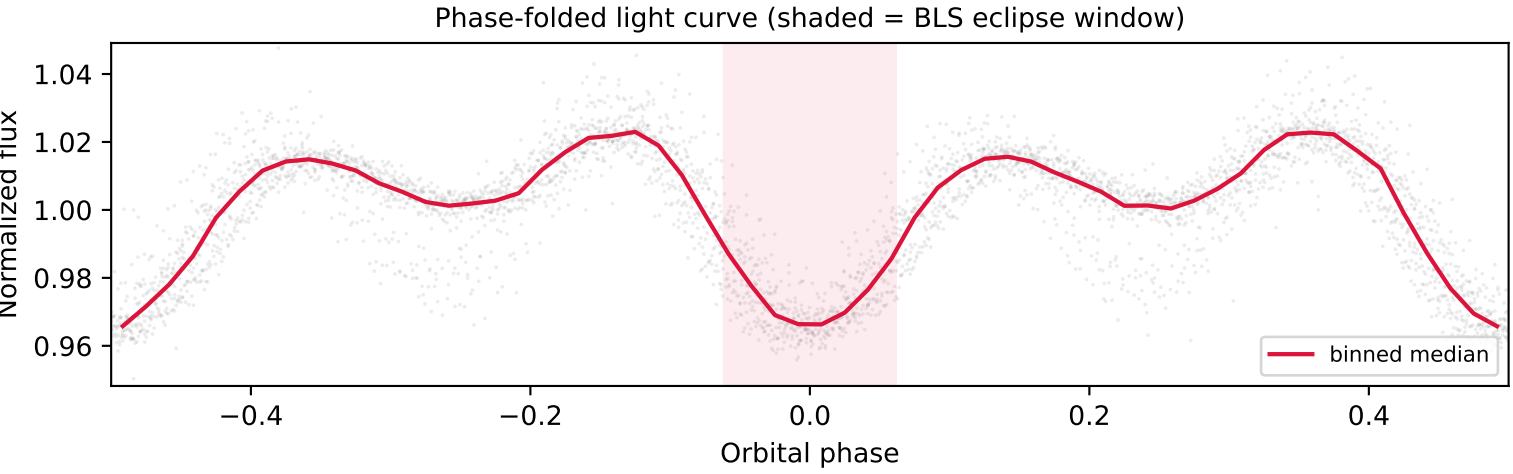
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      61  (gate >= 3)
  Robust depth ratio     1.26  (gate >= 0.50)
  Robust depth sigma     46.6  (gate >= 7.0)
  Odd/even z-score       1.94  (consistent)
  Secondary eclipse      not detected
```

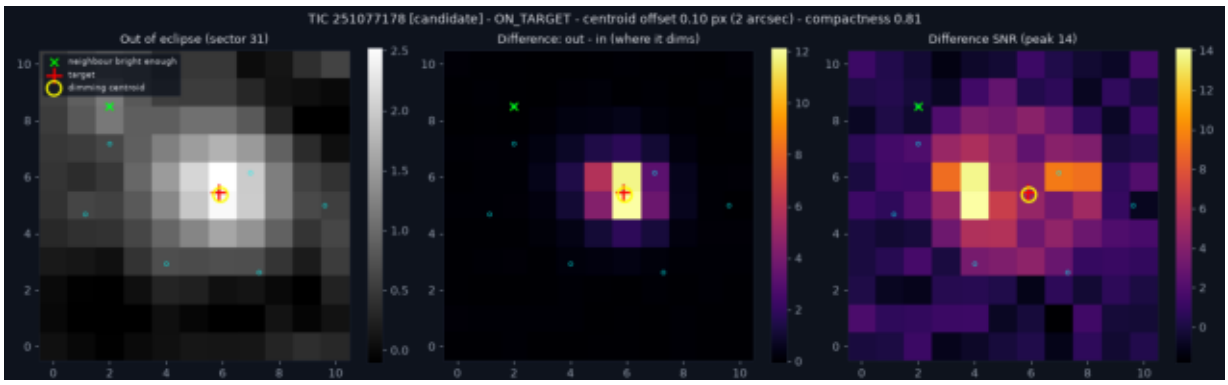
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 31    light-curve dip in this sector: 71.8 sigma    t0 refined by 0.4950 phase
  Difference-image peak SNR   14.17   (gate >= 5.0)      Compactness   0.81   (gate >= 0.50)
  Dimming centroid offset     0.10 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 251077176
    5.02 px away, dTmag +3.52 - farther from the dimming than the target
```





No caveats - clean detection on every check.